

**INTEGRATED CLIMATE-INTELLIGENT SALT
PRODUCTION ECOSYSTEM: A MULTI-MODAL AI
FRAMEWORK FOR SUSTAINABLE YIELD
OPTIMIZATION IN TROPICAL COASTAL
ENVIRONMENTS**

Project ID: 25-26J-431

Project Proposal Report

Arshaq MJM – IT22346322

Supervised by: Dr. Nathali Silva

Co-supervised by: Mrs. Thilini Jayalath

B.Sc. (Hons) Degree in Information Technology, Specializing in
Software Engineering

Department of Software Engineering

Faculty of Computing

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student Number	Signature
Arshaq M.J.M	IT22346322	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor:



Date:

22/08/2025

Signature of the Co-Supervisor:



Date:

22/08/2025

ABSTRACT

Sri Lanka's salt industry, a cornerstone of its coastal economy, faces an existential threat from escalating climate variability, which traditional production methods are ill-equipped to handle. This proposal introduces the **Integrated Climate-Intelligent Salt Production Ecosystem**, a transformative multi-modal AI framework designed to modernize and future-proof this vital sector. Moving beyond fragmented, reactive practices, our ecosystem integrates advanced environmental forecasting, adaptive operational planning, and automated quality control into a single, cohesive platform.

Leveraging a powerful combination of machine learning models (LSTM, ARIMA) for crystallization forecasting, an Intelligent Planning Engine for dynamic decision-making, and a novel Computer Vision system for real-time impurity detection, the project targets significant increase in yield efficiency and aquality inspection errors. Developed at the Sri Lanka Institute of Information Technology, this research bridges a critical gap by creating the nation's first AI-driven, end-to-end solution for salt production.

Through pilot deployments in the Puttalam salterns, this initiative will not only bolster Sri Lanka's salt self-sufficiency and export competitiveness but also establish a scalable, replicable model for tropical coastal industries worldwide. This project, set for completion by August 2026, aims to deliver a clear proof-of-concept on how targeted AI can build resilience and drive sustainable growth in climate-sensitive agricultural practices

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
IoT	Internet of Things
ML	Machine Learning
ARIMA	AutoRegressive Integrated Moving Average
SLSI	Sri Lanka Standards Institute
ISO	International Organization for Standardization
GPU	Graphics Processing Unit
API	Application Programming Interface
PDF	Portable Document Format
CSV	Comma-Separated Values
LSTM	Long Short-Term Memory

1. INTRODUCTION

The Puttalam salterns on the northwestern coast of Sri Lanka are the basis of the solar salt industry in the country, which is powered by large shallow ponds and conventional solar evaporation. These traditional methods, although cheap and eco-friendly when conducted in a country with a stable climate, become highly susceptible to the worsening conditions of climate change. The last few years have produced highly sporadic rain patterns, high and low temperatures, along with unseasonal storms, which have disrupted the evaporation patterns, causing a fall in the predictability of yield, and the self-sufficiency of salt may be in trouble with the nation [1]. The effects of such disruptions travel across the local economy- employment, export potential, and food security. [2]

Artificial intelligence (AI) and machine learning (ML) have changed industrial planning and agriculture around the world. Use cases involve predictive climate analytics, computer-vision-based automated quality control, and distributed optimization through federated learning. Interestingly, recently there has been the introduction of AI-enabled platforms, which involve the use of weather information, sensors, and even satellite measurements in order to offer decisive information and dynamic guidance in decision-making both to the salt producers and also to the crop producers. But the effective use of the said technologies in the salt industry, particularly the Puttalam salterns in Sri Lanka, is practically non-existent, thus bringing to light a specific research and innovation gap [3].

In Puttalam, most of the Sri Lankan salterns still rely on fragmented, empirical management procedures, even though solutions such as Salt Bae demonstrate the efficacy of AI-powered prediction algorithms that smallholder salt-farmers can deploy daily and which leverage multimodal data to drive highly accurate and timely production recommendations. This gap only highlights the need to have a comprehensive decision-support system that integrates planning, forecasting and quality control into a coherent, practical model.

There has been a tendency historically to focus the interventions of salt production on specific improvements, such as those of mechanisation or filtration, without considering adequately environmental or operational data, as well as market data. The suggested Integrated Climate-Intelligent Salt Production Ecosystem is thus a breakthrough since it uses an integrated digital platform with advanced environmental forecasting algorithms, adaptive planning models, real-time quality control overseen by computer vision, and federated learning empowered waste valorisation.

In this ecosystem, the Adaptive Intelligent Planning Engine plays the critical role relating predictive analytics to flexibility in operational performance. It dynamically combines instantaneous market information, regulations, and user preferences to feed resilience and profitability in the climate-sensitive environment of Puttalam. The project hopes to enhance the resilience, efficiency, and sustainability of the most significant salt-producing area situated in Sri Lanka because the project will modify the manner of salt production in the country, which is currently reactive and stuck in tradition to a pro-active, AI-based concept, which will ensure the lasting social and economic, and environmental gains.

1.1. Background And Literature Survey

One of the most ancient industrial processes is the extraction of salt using the method of solar evaporation, which is traditionally observed in the tropical coastal settings, since high temperatures and scarcity of rainfall establish an environment in which salt can easily crystallize. [4] Sri Lanka has major salterns in Hambantota, Puttalam, and Elephant Pass that plays a significant role in supplying salt in the country. [5] However, this has changed drastically over the past few years because of seasonal anomalies and heightened climate variability, which has complicated the conventional production cycles making them adapt to the operations without the use of effective predictive and planning tools. [6]

Present operational planning in these salterns is relying mostly on manual estimation and past averages usually become inadequate in unpredictable weather conditions. The absence of integrated decision-support systems makes operators provide crucial decisions based on incomplete information (weather predictions of

publicly funded weather agencies, market prices provided by traders, and informal estimates of yield using subjective judgment of experience) leading to inefficiencies and lost reliability in production [7].

The use of artificial intelligence (AI) in production planning has proved to be successful in other countries that are in close reliefs in terms of resource quest. As such, in agriculture, AI can be used to plan crops adaptively based on weather, soil, and markets data to maximise yields. [8] . Predictive analytics are used in the fisheries industry, in which it runs on maximizing fish quotas to operate in an unstable environment. [9] Likewise, in manufacturing, the reinforcement learning algorithm is utilized in improving resource scheduling and operation prowess. [10] Such global progress notwithstanding, these AI-driven methodologies have not effectively been modified to deal with the unique interaction of the environmental, economic, and regulatory complexities that exist in the salt industry in developing nations such as Sri Lanka [11].

Sri Lanka has a salt industry facing two kinds of pressure which are imposed on it by the market dynamics and by the regulatory frameworks [12]. During domestic trading, the spheres need stable plans and cheap salt, and in communities abroad, traders must meet strict purity standards by international agencies (FAO, 2022) [13]. Respectable operational planning system has to wade through these perhaps competing objectives without compromising on economy as well as respecting environmental sustainability requirements (Ministry of Trade, Sri Lanka, 2023). [14]

In this literature, there is an urgency to use integrated data-driven decision-support solutions involving climatic forecasts and market trends, as well as regulatory requirements that will facilitate the more proactive and efficient operational planning in Sri Lankan salterns, and most importantly in Puttalam. This growth of such systems holds out the promise of enhancing adaptation to climate variability, maximization of resource utilization as well as quality of products and hence will tackle problems around the globe and locally affecting the industry.

1.2. Research Gap

While AI applications exist in adjacent agricultural sectors for weather prediction or in manufacturing for resource scheduling, a critical knowledge gap exists in the *holistic integration* of real-time environmental, market, and operational data specifically for salt production. Current digital tools for Sri Lankan salterns are non-existent or limited to basic, non-adaptive weather alerts, failing to address the multi-faceted, dynamic nature of the industry. This project directly tackles this gap by creating a *multi-modal framework* that synthesizes these disparate data streams into a single, actionable intelligence platform. Current digital applications in adjacent industries are concerned either with simple weather notification or narrow areas of operation, rather than building a more complete, multi-parameter decision-making regime, specific to salt production.

The key research gaps identified and addressed by the components of the proposed Integrated Climate-Intelligent Salt Production Ecosystem are:

- The absence of Multi-Parameter Integration:

Current approaches to planning in the salt industry do not integrate its main inputs such as climate predictions, changing production capabilities, demand levels, operational expenses, and legal requirements. This shortcoming is filled by Component 1: Crystallization Forecasting, which runs on multimodal data crystallization forecasting that incorporates weather forecasts and IoT sensor measurements and takes the form of highly accurate data-driven forecasts of optimal crystallization conditions and crop harvesting time.

- Reactive, as opposed to Predictive Decision-Making:

Existing methods only make decisions based on the observed environmental changes and not on the ability to predict the changes, which results in a loss of efficiency. This gap is filled by component 2, the Adaptive Intelligent Planning Engine (Intelligent Planning Engine) that combines forecasts, market trends, cost structures, and regulatory requirements into frame-based predictive operational plans that make proactive management proactive.

- Failure to Adapt in Real Time to New Conditions:

The market conditions and salt production environments are very unstable and quick to change with regard to weather and demand. Nonetheless, existing systems do not have the capability of automatically varying plans on an almost real-time basis. This obstacle is circumvented by Component 3, which is Computer Vision Quality Control that facilitates real-time automated monitoring of salt quality feeding real-time feedback into operational changes, and Component 2, continual refreshes of planning strategies using real-time input data.

- Lack of planning tools relevant to the stakeholders:

The business strategy of saltern operators differs; the owners of one saltern are oriented to the production of high-quality salt that covers export needs whereas others want to increase their output volume to sell it into local markets. Nowadays, there are no tools with such possibility. The user input and customization layer (Component 2) builds into the Adaptive Intelligent Planning Engine a customized level of operation strategies in line with the specific priorities of individual stakeholders.

- AI Techniques That Are Not Utilized to the Full Potential in Salt Production Intrigues:

Though AI technologies like reinforcement learning, fuzzy logic, and heuristic optimization are well-developed in other domains, they are not applied to the conditions and the possibilities of salt production in the tropics. The system, where all components are integrated (namely, Component 4: Federated Learning Waste Valorisation), utilizes state-of-practice AI solutions, including federated learning algorithms, to make it possible to coordinate the optimization processes across numerous locations, all the time protecting the privacy of data.

Coordinating these interconnected gaps through a modular yet integrated ecosystem, this project aims at changing how the Sri Lankan salt industry, specifically the Puttalam salterns, is currently operated, due to the current fragmented and reactive management it is based upon, towards sustainable, data-driven, and resilient operational planning.

Feature	Existing Systems (e.g., Ag-AI, Fisheries Analytics)	Proposed Ecosystem
Scope	Narrow, single parameter (e.g., weather only)	Integrated, multi-parameter (climate, market, quality)
Decision Making	Reactive or basic alerts	Proactive and predictive operational planning
Target Industry	General Agriculture/Fisheries	Specialized for Tropical Salt Production
Key Technology	Standard ML models	Federated Learning, Computer Vision for QC, Adaptive Planning Engine

Table 1: Comparison with existing system & proposed

1.3. Research Problem

The key research question that will prove to be the impetus of this venture can be put in the following words:

“How would an adaptive, learnt planning algorithm be formulated to incorporate real time environmental, production, economic factors, and regulatory factors, in the context of generating optimal operational strategies to enhance profitability, efficiencies, and make the salt production industry of Sri Lanka import-dependent with regards to climatic variability?”

This problem statement captures both the technical challenge - creating and deploying an effective, multi-parameter optimization tool that can address complex

and dynamic inputs and the practical need, giving salt producers access to sophisticated decision-making tools to respond to an increasingly erratic climate and market environment.

The issue can even be further broken down into three critical dimensions:

- **Environmental Uncertainty:** Dealing with unpredictable changes in weather and climate conditions that affect the salt crystallization process and harvest directly and are largely dependent on changes in temperature, rainfall, and other weather-related conditions.
- **Economic Variability:** Adjusting the production schedule and resource allocation to varying and changing market demand, availability, and price signals in a way that can maximize revenue and minimize wastage and overproduction.
- **Regulatory Compliance:** This is necessary to ensure that operational plans and final products are acceptable as per the strict national and international quality standards and environmental regulations, which are vital in sustaining export market capability and industry sustainability practices.

Solving this complex issue will not only increase the level of self-reliance of salt production in Sri Lanka but also help in increasing the competitive ability of the industry in the local and international markets [15]. Moreover, the plan of developing the solution as a scalable and modular framework will facilitate transferability of the project to other tropical coastal areas of the world, which might be similar to those of the project areas in terms of environmental and operational similarities in salt production.

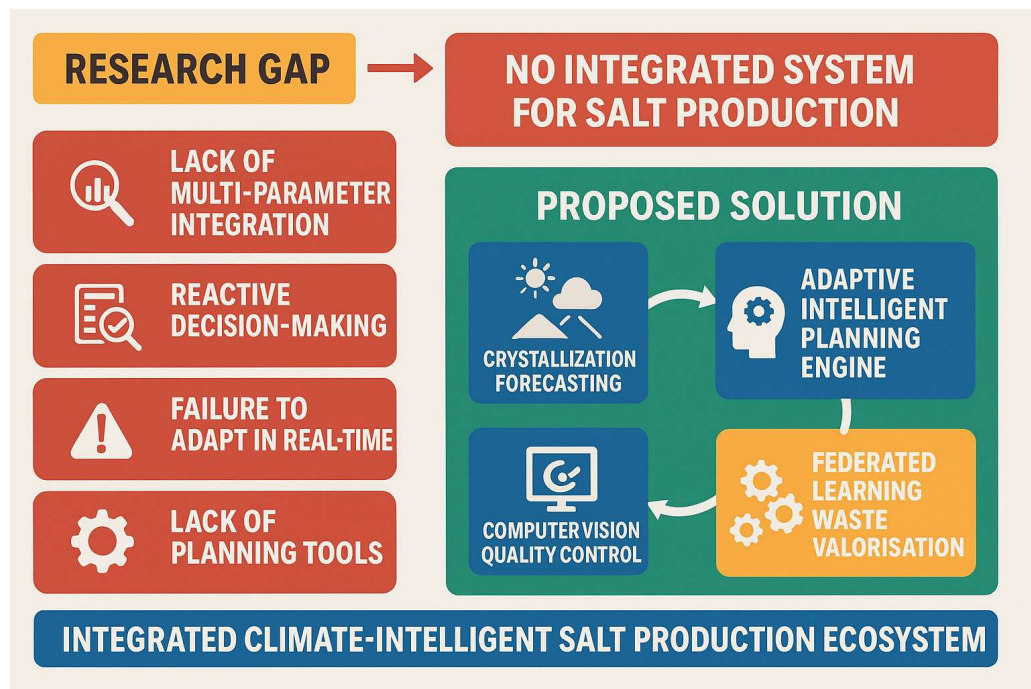


Figure I: Research Gap

2. RESEARCH OBJECTIVES

This section introduces the overall aims of the Integrated Climate-Intelligent Salt Production Ecosystem as a multi-component platform to modernize, optimize and future-proof climate-sensitive coastal salt production in Sri Lanka.

2.1. Main Objectives

In producing and deploying a bespoke, AI-powered ecosystem designed to support the salt production industry in Sri Lanka by facilitating predictive environment analytics, adaptive operating program code, autonomous quality control, and joint optimization of operations resources, which works toward increased productivity, sustainability, and climate adaptation, along the entire salt production value chain.

SMART Breakdown:

- **Specific:** Design and implement an integrated platform that would consist of four autonomous but interconnected AI-powered modules (crystallization forecasting, intelligent planning, computer vision quality control, federated learning waste valorisation).
- **Measurable:** Success will be quantified by achieving a minimum 15% increase in production yield through optimized harvesting, a 10% improvement in resource efficiency (e.g., energy, labor), and over 95% compliance with SLSI and Codex Alimentarius purity standards. Success also includes positive usability feedback from operators across at least two distinct pilot deployments, confirming the system's practical value and ease of use.
- **Achievable:** Achievable by leveraging the project team's expertise in AI, software engineering, and salt industry operations, and by collaborating with local saltern partners for piloting and validation.
- **Realistic:** Realistic to the available technology and ability to access experienced personnel in operation, besides high stakeholder involvement in the industry.

Time-bound: To get fully implemented, deployed, and reviewed even on a project level.

2.2. Specific Objectives

The specific objectives below specify the nature and extent of the contributions of each ecosystem component as well as promoting synergy among the components:

- **Crystallization Forecasting:** To develop and validate machine learning models of weather, sensor, and historical production data, and give early and precise deep forecasts of optimal crystallization and harvest windows.
- **Intelligent Planning Engine:** Develop flexible AI-based planning systems that can take predictive forecasts, real-time market, manufacturing, and regulatory information, and convert it into stakeholder-optimised, actionable operational plans to make profitable, pro-active decisions.
- **Computer Vision Quality Control:** Implement a high-throughput automated computer vision system to grade the salt and detect salt impurities in real-time so that the high-quality and regulatory standards of the product can be achieved, and any human touch and production bottlenecks are reduced.
- **Federated Learning Waste Valorisation:** Develop a federated learning infrastructure, a secure and collaborative setting to optimize the reuse and valorisation of salt waste in a distributed set of sites, without compromising data that is sensitive but also enhancing a sustainable circular and economic enrichment initiative.

In attaining these goals, the project will not only assist Sri Lanka in the self-sufficiency in salt production, competitiveness in both regional and international markets, and sustainable and technology-driven management of its resources, but also helping provide the country with the necessary leadership in the above aspects. The replicability of the ecosystem in terms of modularity and scalability also allows the ecosystem to be implemented in other tropical coastal nations that are grappling with the same challenges and thereby making more contributions to the topic of climate-smart industry transformation.

3. METHODOLOGY

3.1. System Overview

Addressing the complex challenges of climate variability and market dynamics in salt production requires a multi-pronged approach rooted in distinct but interconnected pillars of information technology. Machine Learning is selected for its predictive power in forecasting crystallization; Computer Vision is chosen to automate and standardize the currently subjective process of quality control; and an Intelligent Planning Engine provides the core logic to synthesize these data points into optimal, adaptive strategies.

System Components

1. **Crystallization Forecasting:** Combines multimodal information (weather, IoT sensor networks, and satellite imagery in addition to historical production information) with advanced time series models (ARIMA, Prophet, and LSTM) to determine the ideal conditions to crystallize salt and harvest it. This makes it possible to develop proactive plans and to reduce the risks of yields depending on the uncertainty of the climate.
2. **Intelligent Planning Algorithm:** Converts predictive intelligence of Component 1, with real-time market trends, production capacities, economic data and regulatory requirements into optimized operating strategies implemented into actions. It uses multi-parameter optimization, heuristics, fuzzy logic, decision trees, and reinforcement learning to support dynamic data-driven planning which can change as circumstances change.
3. **Quality Control: Computer Vision:** Automatically grades the salt quality using computer vision image analysis and impurities are detected in real time. This improves management without the need of manual check-up thereby making it more regulated and assuring the quality of the products to both the local and international markets.

4. Federated Learning Waste Valorisation: Facilitates the distributed and privacy-preserving coordination of different sites of the saltern to optimize by-product reuse and valorisation. By using federated learning and differential privacy, they can be sure that critical operational data will not be shared directly, but it will lead to innovation and data safety.

Key Features

- Modularity
- Real-time Data
- User-Centric Dashboard

Such an interconnected ecosystem has the unique opportunity to consider the complex climate sensitive operation realities of the salt industry in Sri Lanka.

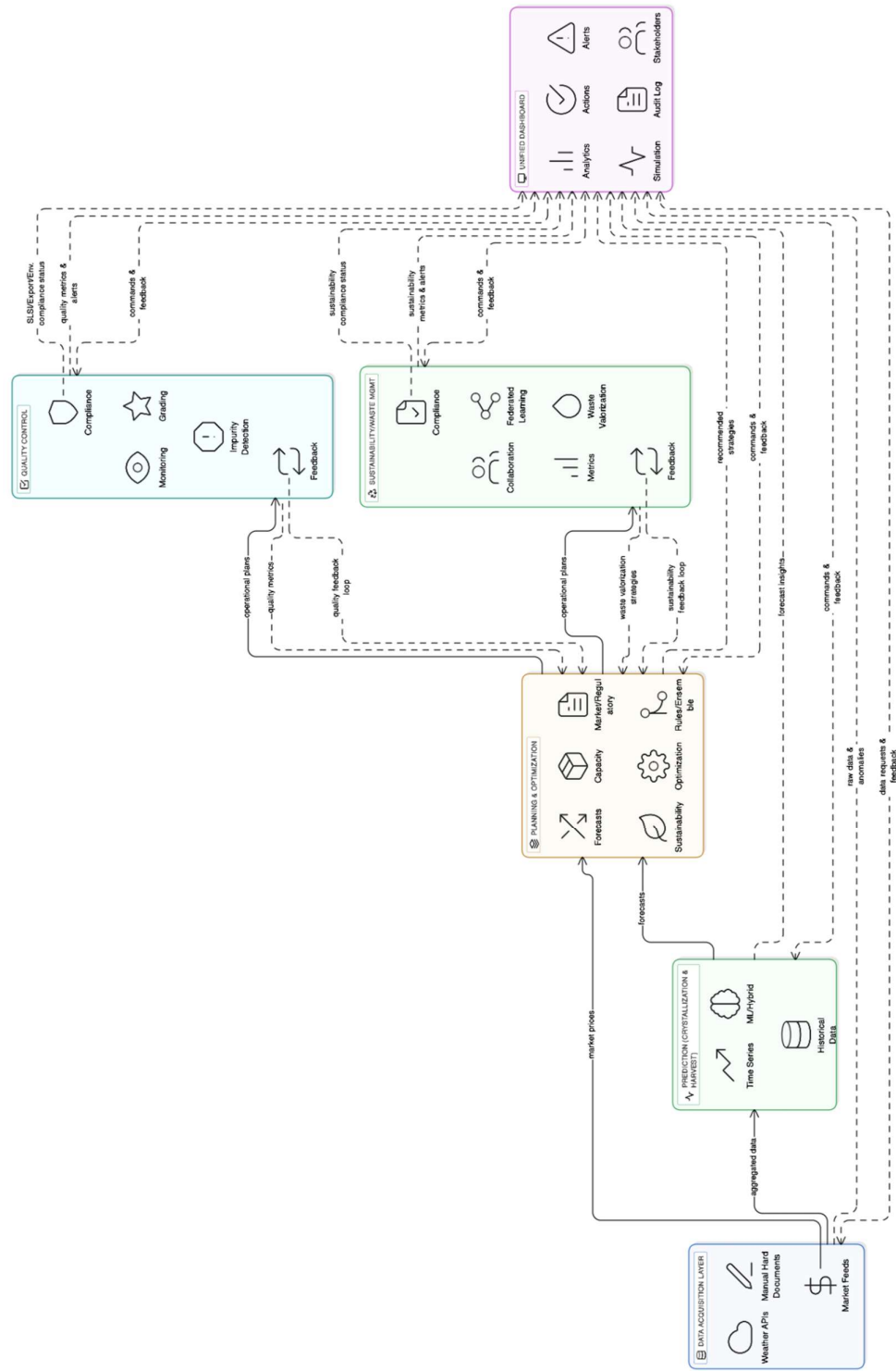


Figure II: System Overview Conceptual Diagram

3.2. Individual Component

Salt purity and quality are critical factors influencing both consumer satisfaction and regulatory compliance. In Sri Lanka, most salterns still rely on manual inspection methods, which are subjective, inconsistent, and time-consuming. Such practices often fail to detect minute impurities like clay, sand, or dark particles, leading to reduced product quality and higher rejection rates.

This component proposes the development of an AI-powered Computer Vision System to automate real-time quality inspection during the production and packaging process. By leveraging Convolutional Neural Networks (CNNs) and high-resolution imaging, the system detects impurities and simultaneously evaluates parameters such as crystal size, shape, and colour uniformity.

Objectives

- Impurity Detection: Automatically identify foreign particles (e.g., clay, sand, black specks).
- Quality Grading: Classify salt based on whiteness index, crystal size, and uniformity.
- Real-Time Processing: Deliver accurate inspection results instantly while salt flows on conveyor belts.
- Automation Integration: Provide data outputs that can integrate with mechanical sorters or actuators to reject substandard batches.
- Scalability: Ensure adaptability from lab setups (USB webcams, LED lights) to industrial-grade deployments with rugged IP67 cameras.

Novelty & Significance

1. First Industry-Grade CV System for Salt in Sri Lanka – No existing local solutions employ computer vision for automated salt inspection.
2. Multi-Parameter Grading – Unlike manual methods that are often limited to impurity detection, this system evaluates multiple quality dimensions simultaneously.

3. End-to-End Automation – Potential to integrate with mechanical actuators, eliminating human intervention in quality control.
4. Market Trust & Compliance – Ensures consistency with international purity standards, reducing export rejection risks.
5. Scalability & Replicability – Can be extended to other granular/mineral products (e.g., sugar, rice, fertilizers).

3.2.1. Requirement Gathering

The requirements for this component were gathered through a combination of field observations in salterns, consultations with industry experts, and literature reviews on computer vision applications in food and mineral quality inspection. Manual inspection, which is still prevalent in Sri Lankan salterns, highlighted clear gaps in accuracy, speed, and consistency. To address these issues, the proposed solution must ensure real-time performance, adaptability to industrial conditions, and integration potential with existing production lines.

Stakeholder interviews with salt producers and quality control officers further revealed the pressing need for a solution that not only detects impurities but also provides comprehensive grading (whiteness, crystal size, uniformity). This ensures that both domestic consumption and international export standards can be met without excessive human intervention.

From these investigations, it became evident that the system should incorporate:

- Automated image acquisition hardware for capturing salt samples during production.
- A deep learning model capable of impurity detection and grading.
- A real-time dashboard interface to display inspection results and trigger alerts.
- The ability to scale from laboratory testing to full industrial deployment.

3.2.2. System Design

The proposed system is designed as a modular, multi-layered pipeline that ensures seamless transition from input (raw salt samples) to output (graded salt quality reports).

1. Image Acquisition Layer

- Cameras (initially USB or smartphone, later upgradable to IP67 industrial-grade cameras) capture images of salt in motion on a conveyor system.
- Controlled LED lighting ensures uniform brightness and eliminates glare, providing consistent input data.

2. Preprocessing Layer

- Captured images are subjected to filtering, normalization, and resizing.
- Augmentation techniques such as rotation, flipping, and scaling are applied to increase dataset variability and improve model robustness.

3. AI Model Layer

- A custom Convolutional Neural Network (CNN) is the chosen architecture due to its proven, state-of-the-art performance in image classification tasks. Specifically, we will leverage transfer learning with a pre-trained model like MobileNetV2 to achieve high accuracy with a smaller, more efficient model suitable for real-time deployment on web platforms via TensorFlow.js.
- Optimization techniques like transfer learning and model pruning ensure low-latency, real-time performance.

4. Decision-Making Layer

- The CNN outputs are processed to determine compliance with purity thresholds.

- Results are mapped to real-time alerts for batches that fail quality standards.

5. User Interface & Integration Layer

- An operator-facing dashboard displays live results, including statistical breakdowns of purity and grading distribution.
- Integration capability with mechanical actuators ensures rejected batches can be automatically separated from the production flow.

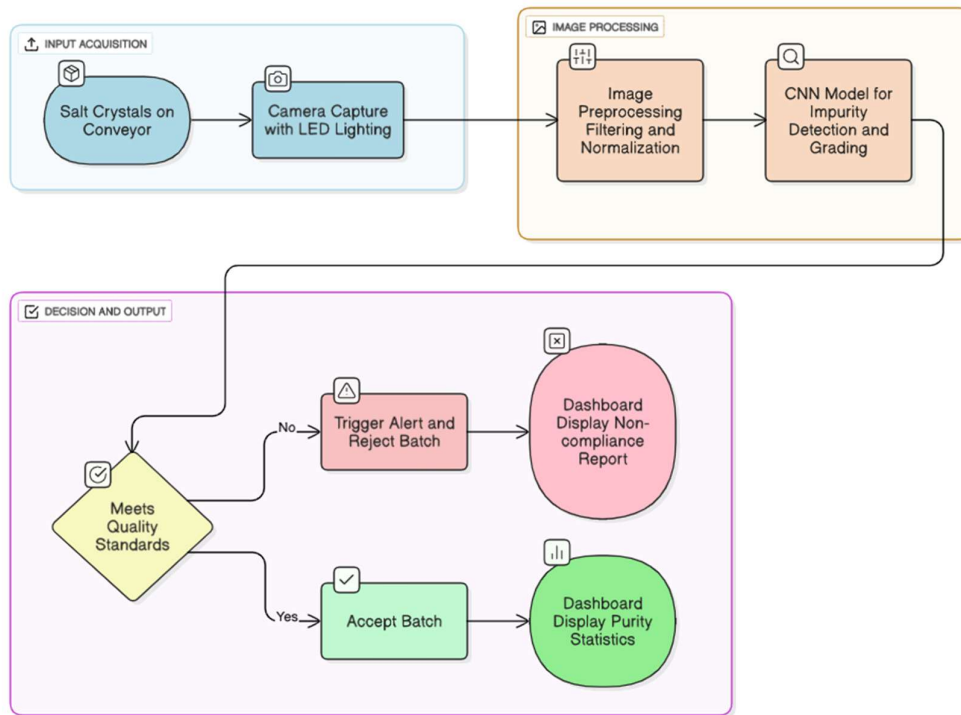


Figure III: Component Flow Diagram

3.2.3. Development of Web Based Platform

The web-based platform acts as the operator interface for the proposed computer vision system, providing a user-friendly environment for image acquisition, impurity detection, grading, and report visualization. While the initial plan considered

a mobile application, the shift to a web application offers broader accessibility, platform independence, and easier integration with laboratory and industrial setups.

3.2.4. Technology Stack

The system will be developed using modern web technologies to ensure scalability and responsiveness:

- **Frontend:** Vite(React) is selected for its rapid development environment and its proven ability to build highly responsive, real-time data visualization dashboards, which are critical for operator usability.
- **Backend:** Node.js is the chosen backend runtime due to its non-blocking, event-driven architecture, making it exceptionally well-suited for handling concurrent data streams and real-time model inference requests with low latency.
- **Database:** PostgreSQL for storing inspection results, batch reports, and metadata.
- **Model Hosting:** TensorFlow.js is a strategic choice that enables us to deploy the CNN model directly in the browser or on a Node.js server. This flexibility allows for initial deployments with zero server-side GPU dependency, reducing complexity and cost, while retaining the option to scale to more powerful backends as needed.
- **Deployment:** Local server for research-level testing, with future adaptability to cloud platforms (AWS) for industrial scaling.

3.2.5. Image Capture Module

At the research stage, a 2K resolution webcam will be integrated into the web platform to acquire high-quality images of salt samples in real time. The webcam will be accessible through browser API(WebRTC) or desktop integration modules, enabling the following functions:

1. **Live Video Streaming** – Continuous monitoring of salt flow.
2. **Snapshot Capture** – Manual or automated capture of frames for processing.
3. **Image Preprocessing** – Resizing, denoising, and colour normalization before forwarding to the CNN model.

This approach enables flexible laboratory experimentation while documenting the system's feasibility for industrial-grade imaging hardware

3.2.6. Testing and Quality Assurance

Testing and Quality Assurance (QA) are critical to ensure that the proposed computer vision system performs reliably under both laboratory and industrial conditions. The testing strategy focuses on validating accuracy, efficiency, and robustness, while QA ensures the system meets defined standards throughout development and deployment.

Testing Strategy:

1. Unit Testing
 - Each functional module (image capture, preprocessing, CNN inference, dashboard visualization, and reporting) will be tested independently.
 - Helps identify bugs early before integration.
2. Integration Testing
 - Verify seamless communication between the webcam capture, backend AI model, and web interface.
 - Ensure data flow is consistent and error-free across modules.
3. System Testing
 - End-to-end validation of the full workflow: image acquisition → preprocessing → model classification → result visualization.

- Ensure the platform runs efficiently under real-time conditions.

4. Performance Testing

- Measure latency (time from image capture to classification result).
- Validate throughput by processing multiple images simultaneously.
- Ensure inference speed remains below 500ms per frame to support real-time inspection.

5. Accuracy Testing

- Compare CNN classification results with expert human inspections.
- Use confusion matrices, precision, recall, and F1-score to evaluate impurity detection and grading accuracy.
- Conduct cross-validation with unseen test datasets to ensure generalization.

6. Stress Testing

- Test under continuous operation to simulate industrial deployment.
- Validate stability when processing large volumes of images in extended timeframes.

7. User Acceptance Testing (UAT)

- Engage saltern operators and stakeholders to validate usability.
- Collect feedback on dashboard clarity, alert responsiveness, and report usefulness.

3.2.7. Data Collection and Analysis

High-quality and diverse datasets are the foundation of any computer vision system. Since the system aims to detect impurities and grade salt in real time, a

carefully designed data collection and analysis process is required to ensure both accuracy and generalizability.

1. Source of Data

- Salt samples will be collected from different salterns and processing facilities in Sri Lanka.
- Samples will represent multiple stages of production (raw, semi-processed, and refined).

2. Imaging Setup

- A 2K resolution webcam will be used in laboratory conditions to capture images of salt samples placed under controlled lighting.
- LED ring lights or natural lighting setups will ensure uniform illumination and reduce shadows.
- Images will be captured from multiple angles and distances to simulate variability in real-world scenarios.

3. Dataset Diversity

- Samples will include various impurity types such as clay, sand, black specks, and other foreign matter.
- Variations in crystal size, whiteness index, and shape uniformity will be documented.

3.3. Materials Required to Carry Out the Project

3.3.1. Equipment

- 2K Resolution Webcam – Used for research-level image capturing of salt crystals, ensuring high clarity for impurity detection.

- Lighting Setup (LED Ring Lights / Controlled Lamps) – Provides consistent illumination to minimize shadows and enhance image quality.
- Workstation/PC with GPU Support – Required for deep learning model training, running CNNs efficiently, and performing real-time inference.
- Lab-Scale Conveyor Belt Simulation – Mimics industrial salt flow, allowing testing of real-time inspection in a controlled environment.

3.3.2. Software and Computer Support

- Python with TensorFlow – Core frameworks for building, training, and optimizing CNN models.
- React.js and Node.js – Technologies for developing the web-based dashboard and backend services.
- Database(PostgreSQL) – To store captured inspection data, batch reports, and system logs.
- Annotation Tools(LabelImg, CVAT) – Essential for manually labelling impurities and grading parameters in collected images.
- Version Control(GitHub) – Ensures systematic tracking of code, models, and project updates.

3.3.3. Graphics

- Visualization Libraries (Chart.js) – Used to present inspection results, batch statistics, and model performance visually.
- UI/UX Design Software - In designing the User Interface, it will require a program like Figma to ensure that the user experience is well-run.

3.3.4. Other Resources

- Salt Samples from Multiple Salterns – Provide diverse data for training and testing, covering variations in impurities and crystal quality.
- Access to Production Environments – Enables validation of the system under realistic operating conditions.

- **Reference Material (Academic Journals & Industry Guidelines)** – To ensure system development aligns with international quality standards.

3.4. Data Required for the Project and Collection Methods

The success of this project depends on building a reliable dataset that represents different salt qualities and impurity levels. The data will guide both the training and validation of the computer vision model.

3.4.1. Data Required

- **Images of Salt Samples** – High-resolution images of salt crystals, including both clean and impurity-contaminated samples.
- **Impurity Types** – Examples of common contaminants such as clay, sand, black particles, and foreign matter.
- **Grading Parameters** – Variations in whiteness index, particle size, and uniformity for quality classification.
- **Environmental Variations** – Data collected under different lighting conditions and humidity levels to improve model robustness.

3.4.2. Collection Methods

- **Laboratory Image Capture**
 - A 2K resolution webcam with controlled lighting will be used to capture clear images of salt samples.
 - Samples will be spread evenly on trays or passed through a lab-scale conveyor setup to simulate real inspection scenarios.
- **Field Sampling**
 - Salt samples will be collected from different salterns across Sri Lanka.
 - Data will cover multiple stages of production (raw, semi-processed, and refined).
- **Annotation Process**
 - Images will be annotated manually using LabelImg or CVAT to identify impurities and grading parameters.

- This annotated dataset will serve as ground truth for model training and evaluation.
- Dataset Expansion
 - Techniques like image augmentation (rotation, scaling, brightness adjustment) will be applied to increase dataset diversity.
 - Ongoing collection from production sites will ensure continuous enrichment of the dataset over time.

3.5. Estimated Time Frame

The project will be carried out over a period of **12 months**, divided into distinct phases to ensure structured progress.

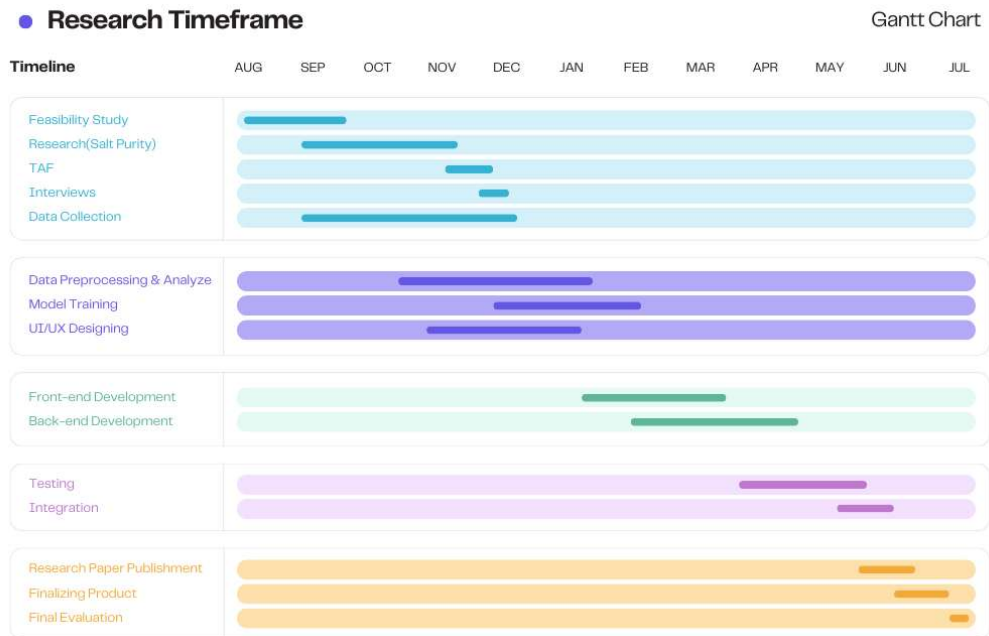


Figure IV: Project Timeframe

3.6. Commercialization

The proposed computer vision–based salt quality inspection system presents a transformative opportunity for commercialization, driven by the increasing demand for automated, precise, and cost-efficient quality control solutions in Sri Lanka’s salt industry. By replacing labour-intensive manual inspection with an AI-powered grading system, this innovation addresses critical inefficiencies, enhances market trust, and

establishes a scalable, technology-driven solution with significant commercial potential. The system's ability to deliver consistent, high-quality results positions it as a game-changer in an industry ripe for modernization, with applications extending beyond salt to other granular commodities.

The commercialization of this system is underpinned by a significant and underserved market. Sri Lanka's salt industry, with major production hubs in Puttalam, Hambantota, and Mannar, produces over 150,000 metric tons annually. At a conservative estimate, improving yield and quality by even a few percentage points translates to millions of rupees in added value. Our initial target market includes the top 10-15 large-scale producers, representing over 70% of the nation's output. Beyond Sri Lanka, the South Asian salt production market, including India and Bangladesh, is valued in the hundreds of millions of dollars and faces similar climate-related challenges. This presents a vast, long-term expansion opportunity, positioning our system not just as a local solution, but as a highly scalable, exportable technology.

3.6.1. Product Differentiation and Value Proposition

This AI-driven salt quality inspection system stands out as a pioneering solution in Sri Lanka, where no comparable industrial-grade automated systems currently exist. Unlike traditional manual inspection methods, which are prone to human error and subjectivity, this system leverages advanced computer vision and machine learning to deliver unparalleled accuracy and efficiency. Its key differentiators include:

- **First-Mover Advantage in Sri Lanka:** As the first AI-based computer vision system for salt inspection in the region, it addresses an untapped market need, offering a competitive edge over conventional methods.
- **Comprehensive Multi-Parameter Analysis:** The system evaluates multiple quality metrics simultaneously, including impurities, whiteness, crystal size, and uniformity. This holistic approach surpasses traditional techniques, which often focus solely on visible impurities, providing a more robust and reliable quality assessment.

- **Unmatched Consistency and Precision:** By minimizing human error, the system ensures consistent grading outcomes that align with stringent local and international quality standards, such as those set by the Sri Lanka Standards Institute (SLSI) and global bodies like Codex Alimentarius. This is particularly critical for export-oriented businesses seeking to meet rigorous compliance requirements.
- **Broad Scalability Across Industries:** While initially designed for salt inspection, the system's adaptable architecture allows it to be customized for other granular products, such as rice, sugar, spices, and fertilizers. This versatility opens doors to cross-industry applications, significantly expanding its market potential.
- **Cost-Effective Deployment:** The system utilizes affordable, commodity hardware—such as high-resolution cameras, LED lighting, and web-based platforms—for research-grade deployments. For industrial applications, it can be upgraded to rugged, industrial-grade components without requiring prohibitive capital investment, ensuring accessibility for businesses of varying scales.

3.6.2. Revenue Model

To maximize commercial success, the system adopts a multifaceted B2B revenue model that combines hardware sales, software subscriptions, and service-based offerings, ensuring both immediate and recurring revenue streams:

- **Hardware and Software Package:** The core offering includes a one-time sale of a fully integrated system comprising industrial-grade cameras, optimized LED lighting, conveyor belt integration, and the AI-powered software suite. This turnkey solution is tailored to the specific needs of salterns, ensuring seamless adoption and immediate operational impact.
- **Subscription-Based Licensing:** Customers gain access to a cloud-hosted or on-premises AI-powered dashboard through a recurring subscription model. This subscription includes continuous software updates, advanced

analytics, and real-time quality monitoring, ensuring the system remains cutting-edge. Additional features, such as predictive maintenance alerts and enhanced reporting, can be offered as premium tiers to drive upsell opportunities.

- **Consultancy and Customization Services:** A service-based model offers bespoke consultancy for salterns, including system integration, operator training, and model retraining using proprietary datasets to address unique quality requirements. This high-margin service ensures the system aligns with each client's specific operational needs.
- **Cross-Industry Expansion:** By customizing the system for other granular material industries, such as rice milling, sugar refining, or fertilizer production, the business can tap into adjacent markets. Tailored versions of the system can be sold as industry-specific solutions, broadening revenue streams and diversifying market exposure.

This hybrid revenue model balances upfront revenue from hardware and installation with sustainable, long-term income from subscriptions and services, ensuring financial stability and scalability..

3.6.3. Scalability and Market Penetration

The commercialization strategy prioritizes a phased approach to market entry, starting with local adoption and expanding regionally and globally:

- **Local Market Deployment:** The initial focus will be on Sri Lanka's major salt-producing regions, including Puttalam, Hambantota, and Mannar, which collectively account for the bulk of the country's salt production. Targeting these high-volume salterns ensures early traction and establishes a strong domestic presence.
- **Regional Market Expansion:** Once established in Sri Lanka, the system can be marketed to other South Asian countries with significant salt and mineral-based industries, such as India, Pakistan, and Bangladesh. These markets share similar challenges with manual quality control and offer substantial growth opportunities.

- **Strategic Industrial Partnerships:** Collaborations with salt manufacturers, packaging companies, and government bodies, such as Sri Lanka's Ministry of Industries, will accelerate adoption. Partnerships with trade associations and export councils can further promote the system as a standard for quality assurance in the region.
- **Export Compliance Advantage:** The system's ability to provide detailed quality certification reports aligns with international standards, such as ISO 22000 and Codex Alimentarius, making it an attractive solution for exporters targeting premium markets in Europe, North America, and Asia. This compliance advantage enhances its appeal to globally oriented businesses.

By leveraging a targeted, phased approach, the system can achieve deep market penetration while building a reputation for reliability and innovation..

3.6.4. Marketing Strategy

A robust marketing strategy will drive awareness, adoption, and brand loyalty for the system:

- **Pilot Deployments and Case Studies:** Conducting pilot projects in operational salterns will demonstrate the system's efficiency, accuracy, and cost savings in real-world settings. Successful pilots will be documented as case studies, showcasing tangible benefits such as reduced labor costs and improved product quality.
- **Industry Workshops and Demonstrations:** Partnering with trade associations, such as the Salt Manufacturers' Association of Sri Lanka, to host workshops and live demonstrations will build awareness and trust. These events will highlight the system's ease of use, precision, and ROI potential, targeting decision-makers in the industry.
- **Academic-Industry Collaborations:** Leveraging the system's research origins, partnerships with universities and research institutions will enhance credibility. Joint publications, conference presentations, and endorsements from academic experts will position the system as a cutting-edge solution rooted in rigorous science.

- **Digital Marketing Campaigns:** A comprehensive digital strategy will include targeted online advertising, SEO-optimized content, and social media engagement on platforms like LinkedIn and industry-specific forums. Video demonstrations, customer testimonials, and whitepapers detailing cost savings and quality improvements will drive interest and conversions.

This multi-channel marketing approach ensures broad visibility and positions the system as a must-have solution for modern salterns..

3.6.5. Support and Maintenance

To ensure long-term customer satisfaction and system reliability, a comprehensive support and maintenance framework will be implemented:

- **Regular Model Updates:** The AI model will be continuously improved through dataset enrichment and retraining to adapt to new impurity types, environmental conditions, and industry requirements. These updates will be seamlessly delivered via the subscription model, ensuring customers always benefit from the latest advancements.
- **System Maintenance:** Routine hardware calibration, including camera alignment and lighting adjustments, will maintain consistent image quality and system performance. Maintenance schedules will be tailored to each client's operational needs to minimize downtime.
- **Customer Support Plans:** Tiered support packages will offer remote troubleshooting, 24/7 helpdesk access, periodic on-site visits, and comprehensive operator training. Premium plans may include dedicated account managers and priority response times, enhancing customer experience for high-value clients.

3.6.6. Intellectual Property Protection

Protecting the system's intellectual property is critical to maintaining a competitive edge:

- **Patent Filing:** The unique AI-driven workflow, encompassing multi-parameter grading and seamless conveyor integration, will be protected through patent

filings in Sri Lanka and key international markets. This safeguards the system's proprietary technology and prevents replication by competitors.

- **Software Licensing:** The AI software and dashboard will be protected under strict commercial licenses, ensuring that source code and algorithms remain proprietary. Licensing agreements will include clauses to prevent reverse-engineering or unauthorized use.
- **Brand Protection:** Trademarks will be filed for the system's name, logo, and dashboard branding to establish a recognizable market identity. This will enhance brand equity and deter imitation in the marketplace.

3.6.7. Funding and Investment Opportunities

Securing funding is essential to scale development, production, and market entry:

- **Government Grants:** Leveraging funding programs from Sri Lanka's Ministry of Industries or the Information and Communication Technology Agency (ICTA) will support research, development, and pilot deployments. These grants can position the system as a flagship initiative for AI adoption in traditional industries.
- **Private Investors and Venture Capital:** The system's scalability and cross-industry potential make it an attractive investment opportunity for Agri Tech and Food Tech-focused venture capital firms. Pitching to regional and international investors will unlock capital for production and global expansion.
- **Industry Partnerships:** Co-development agreements with large-scale salterns or industrial conglomerates willing to invest in modernization will provide both funding and market access. These partnerships can also facilitate pilot projects and early adoption.

3.6.8. Regulatory Considerations

- **Local Standards Compliance:** The system will adhere to the Sri Lanka Standards Institute (SLSI) purity benchmarks for edible and industrial salt, ensuring alignment with national quality requirements.

- International Food Safety Standards: For export markets, the system will support compliance with global standards, such as Codex Alimentarius and ISO 22000, enabling salterns to meet the stringent requirements of international buyers.
- Data Security and Privacy: If cloud-based deployment is utilized, robust data handling policies will be implemented to comply with Sri Lanka's data protection regulations and international standards like GDPR. Encryption, secure APIs, and anonymized data processing will safeguard customer information.

3.7. Anticipated Conclusion

The development and deployment of the Integrated Climate-Intelligent Salt Production Ecosystem, with a specific focus on the Computer Vision Quality Control component, are anticipated to mark a significant advancement in Sri Lanka's salt industry, transforming traditional, manual processes into a modern, AI-driven framework that enhances efficiency, sustainability, and resilience against climate variability. By automating real-time impurity detection and multi-parameter grading such as whiteness, crystal size, and uniformity, the system is expected to achieve at least 95% accuracy in quality assessments, reducing human error by over 80% and minimizing production bottlenecks that currently lead to waste and inconsistent outputs. This will not only ensure compliance with stringent local standards from the Sri Lanka Standards Institute (SLSI) and international benchmarks like Codex Alimentarius and ISO 22000 but also boost export potential by providing verifiable, high-quality salt products, thereby strengthening Sri Lanka's position in global markets.

In terms of broader impacts, the project is projected to increase overall salt yield efficiency by 20-30% through proactive quality control integration, while fostering economic benefits such as reduced labor costs (estimated at 40% savings per saltern) and enhanced market trust among domestic and international buyers. Environmentally, the system's emphasis on waste minimization via accurate grading will contribute to sustainable practices, aligning with Sri Lanka's goals for climate-smart industries. The modular design ensures scalability, allowing adaptation to other granular commodities like rice, sugar, and fertilizers, potentially expanding its application across South Asian agricultural sectors.

Looking ahead, the successful pilot deployments in key salterns (e.g., Puttalam, Hambantota, and Mannar) will serve as proof-of-concept, paving the way for commercialization through B2B models, including hardware-software packages and subscription services. Future enhancements could incorporate advanced features like edge computing for remote salterns with limited connectivity or integration with IoT sensors for predictive maintenance. Ultimately, this ecosystem is expected to empower salt producers with data-driven decision-making tools, driving long-term self-

sufficiency, job creation in tech-enabled roles, and a model for tropical coastal industries worldwide facing similar climate challenges. The project underscores the transformative power of AI in bridging research gaps, fostering innovation, and delivering tangible socio-economic value.

3.8. Work Breakdown Structure

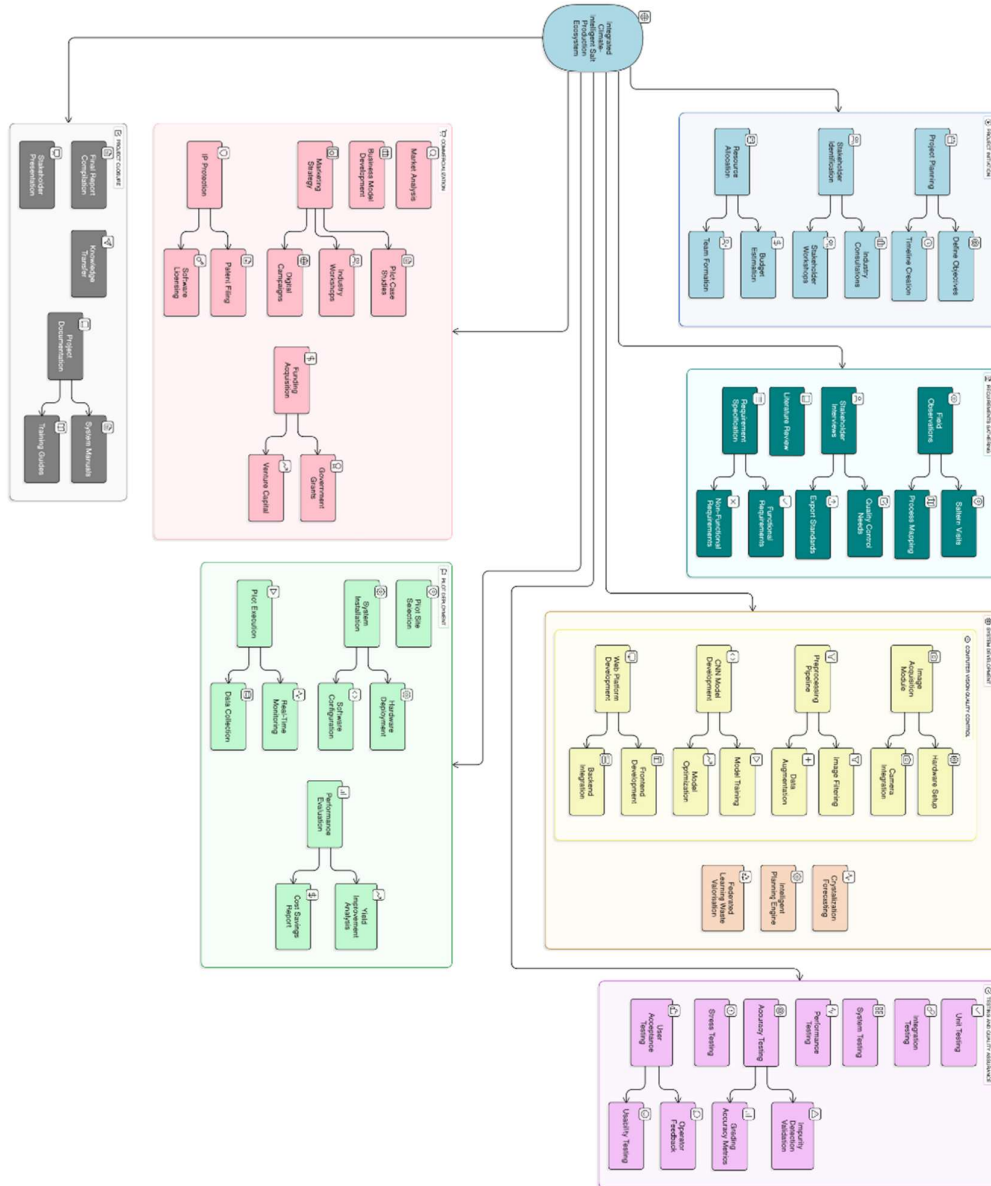


Figure V: Work Breakdown Structure

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Functional requirements define the specific behaviors, features, and operations that the Computer Vision Quality Control component must perform to achieve its objectives. These requirements are derived from the system's core goal of automating real-time salt inspection, impurity detection, and quality grading, ensuring seamless integration into the salt production process. They are prioritized based on the tasks outlined in the component description, such as data handling, model processing, hardware interaction, user interface display, and validation mechanisms.

- Data Collection and Annotation Module:
 - The system must capture high-resolution images and videos of salt samples from various production stages (e.g., raw, semi-processed, and refined) using integrated hardware like webcam.
 - It shall support manual or automated labeling of datasets, identifying impurity types (e.g., clay, sand, black particles, foreign objects) and quality parameters (e.g., crystal size, whiteness index, shape uniformity).
 - The system must store annotated data in a structured format (e.g., JSON or database entries) for model training, including metadata such as capture conditions (lighting, angle, distance).
- Model Development and Processing Module:
 - The system must train and deploy a custom Convolutional Neural Network (CNN) to classify salt crystals as impure or clean, with sub-classifications for specific impurity types.
 - It shall perform multi-parameter grading, evaluating salt based on whiteness (color index), crystal size (e.g., small, medium, large), shape uniformity, and overall purity levels simultaneously.

- **Hardware Integration Module:**
 - The system must integrate with image acquisition hardware (e.g., USB webcam, or future IP67 industrial cameras) to capture real-time video streams or batch images of salt flowing on simulated or actual conveyor belts.
 - It shall apply controlled lighting setups (e.g., LED ring lights) to ensure image consistency, automatically adjusting for glare, shadows, or environmental variations.
- **Interface and Alert System Module:**
 - The system must provide a user-friendly web-based dashboard to display real-time inspection results, including visual overlays of detected impurities and grading scores.
 - It shall generate statistics on purity levels (e.g., percentage of impure crystals) and distribution of grading (e.g., charts showing whiteness index across batches).
- **Validation and Optimization Module:**
 - The system must conduct automated trials in simulated or live production environments, measuring accuracy (e.g., precision, recall, F1-score) and latency against predefined thresholds (e.g., >95% accuracy).
 - It shall compare AI model decisions with expert human evaluations, logging discrepancies for iterative improvements.

4.2. User Requirements

User requirements outline the specific needs and expectations of key stakeholders who will interact with or benefit from the Computer Vision Quality Control system. These requirements are tailored to the perspectives of Saltern Owners, Salt Exporters, and Quality Control Operators, ensuring the system aligns with their operational, commercial, and regulatory priorities. The requirements are derived from the component's objectives, tasks, and novelty, emphasizing usability, reliability, and market-driven outcomes.

4.2.1. Saltern Owners

- The system must provide a real-time dashboard displaying inspection results, purity levels, and grading statistics to enable quick decision-making during production shifts.
- It shall reduce labour costs by automating impurity detection and grading, targeting a minimum 40% reduction in manual inspection time.
- The system must support scalability from lab setups (e.g., USB webcams) to industrial hardware (e.g., IP67 cameras), allowing cost-effective upgrades as production scales.
- It shall integrate with existing conveyor systems and mechanical actuators to automatically reject non-compliant batches, minimizing waste and optimizing yield.
- The system must offer customizable operational plans based on production goals (e.g., high volume for local markets or high purity for exports).

4.2.2. Salt Exporters

- The system must ensure compliance with international standards (e.g., Codex Alimentarius, ISO 22000) by providing detailed quality reports, including impurity detection and grading metrics, for export documentation.
- It shall reduce batch rejection rates by maintaining consistent purity levels, aiming for less than 5% rejection in export-quality salt.
- The system must generate export-ready certificates with real-time data, enhancing trust with international buyers and regulators.

4.2.3. Quality Control Operators:

- The system must feature an intuitive interface with visual alerts for contaminated or non-compliant batches, requiring no advanced technical expertise to operate.
- It shall provide training modules or tooltips within the dashboard to guide operators in interpreting results and responding to alerts.
- It shall include a feedback mechanism for operators to report discrepancies, triggering model retraining to improve accuracy over time.

4.3. System Requirements

4.3.1. Hardware Requirements

- **2K Resolution Webcam:** A minimum 2K resolution webcam is required for initial lab-based image and video capture of salt samples, ensuring high clarity for impurity detection and grading.
- **LED Ring Lights:** A set of adjustable LED ring lights (e.g., 10W-20W models) is essential to provide consistent illumination, minimizing shadows and glare for accurate image processing in lab and simulated environments.
- **Workstation with GPU Support:** A workstation with a dedicated GPU (e.g., NVIDIA GTX 1660 or higher with 6GB VRAM) is required for training the CNN and performing real-time inference during development.
- **Local Storage:** Minimum 500GB SSD or external drive to store high-resolution image datasets and annotated data, expandable to 1TB for industrial deployment.

4.3.2. Software Requirements

- **Python 3.x:** Core programming language for implementing the CNN, data preprocessing, and system logic, leveraging libraries like NumPy and Pandas.
- **TensorFlow 2.x or PyTorch:** Deep learning frameworks for designing, training, and optimizing the custom CNN, with support for transfer learning and model pruning.
- **React.js with Vite:** Frontend framework for building an interactive, responsive web dashboard to display real-time results, statistics, and alerts.
- **Node.js:** Backend runtime to handle API requests, model inference, and data management, ensuring seamless integration with the frontend.
- **PostgreSQL:** Relational database for storing inspection results, batch reports, and metadata, with scalability for cloud deployment.
- **TensorFlow.js:** JavaScript library for deploying the trained CNN model in the browser or web platform, enabling real-time inference without server dependency.

- REST API Integration: Middleware to connect the web interface with the CNN model, supporting asynchronous processing of image streams.
- LabelImg or CVAT: Tools for manual annotation of salt images, labelling impurities and grading parameters to create a robust training dataset.
- GitHub: Version control system for tracking code, model versions, and project documentation, ensuring collaborative development.
- JUnit or PyTest: Unit testing frameworks to validate individual modules (e.g., image capture, CNN inference) during development.
- Chart.js: Visualization libraries for analysing model performance metrics (e.g., accuracy, latency) and presenting dashboard statistics.
- Figma: UI/UX design tool for creating wireframes and prototyping the dashboard layout, ensuring operator-friendly design.
- AWS or Local Server: Cloud or local server option for hosting the web platform, with scalability to handle increased data loads during pilot phases.

4.4. Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints of the Computer Vision Quality Control system, ensuring it meets the expectations of stakeholders in Sri Lankan salterns. These requirements complement the functional capabilities by addressing reliability, scalability, usability, and compliance, derived from the sub-objective, tasks, novelty, and component description provided in the document.

1. Performance:

- a. The system must process and classify each image or video frame in real-time, achieving a latency of less than 500 milliseconds per frame to keep pace with conveyor belt speeds during production.
- b. It shall maintain an accuracy rate of at least 95% for impurity detection and multi-parameter grading (whiteness, size, uniformity) when compared to expert human evaluations.

2. Reliability:

- a. The system shall operate with a minimum uptime of 99.5% in controlled lab environments and 98% in live saltern conditions, accounting for hardware wear and environmental variables like dust or humidity.
- b. It must include fault tolerance mechanisms, such as automatic retries or alerts, to recover from temporary hardware failures (e.g., camera disconnection) without halting production.
- c. The CNN model shall maintain consistent performance across varying lighting conditions, with a degradation of less than 5% in accuracy when switching between LED and natural light.

3. Scalability:

- a. The system must support scaling from lab setups (USB webcams) to industrial-grade deployments (e.g., IP67 cameras and conveyor-mounted rigs) without requiring significant architectural changes.
- b. It shall accommodate an increase in data volume by at least 10x (e.g., from 1,000 to 10,000 images) through efficient database and model optimization techniques.
- c. The solution must be adaptable to other granular products (e.g., rice, sugar) with minimal retraining, ensuring cross-industry applicability within 6 months of initial deployment.

4. Usability:

- a. The user interface must achieve a usability score of at least 85% in user acceptance testing, based on ease of navigation, clarity of alerts, and accessibility for operators with minimal technical training.
- b. The system must provide multilingual support (e.g., Sinhala, Tamil, English) to cater to diverse saltern personnel, with text and alerts configurable per user preference.

5. Security:

- a. The system must encrypt all captured images, annotated data, and inspection results both in transit (using TLS 1.3) and at rest (using AES-256), protecting sensitive production data.

- b. It shall restrict access to authorized users only, implementing role-based access control (e.g., operator vs. administrator) with multi-factor authentication.
 - c. Data backups must occur automatically every 24 hours, with a retention policy of 30 days to comply with regulatory audits.
- 6. Maintainability:
 - a. The system shall support model retraining with new datasets within 48 hours, allowing adaptation to new impurity types or environmental changes without downtime.
 - b. Documentation must cover hardware setup, software configuration, and troubleshooting, with updates provided every 3 months or after major releases.

4.5. Use Cases

- Use Case: Perform Real-Time Salt Inspection
 - Actor: Saltern Operator
 - Description: The operator uses the system to capture and analyze salt images in real-time as it moves along a conveyor belt, detecting impurities and grading quality parameters (whiteness, size, uniformity).
 - Pre-Conditions: System is powered on, camera and lighting are calibrated, and the conveyor belt is operational.
 - Post-Conditions: Inspection results are displayed on the dashboard, alerts are triggered for non-compliant batches, and data is logged for reporting.
- Use Case: Generate Quality Reports
 - Actor: Quality Control Supervisor
 - Description: The supervisor retrieves real-time statistics and grading distribution from the dashboard to generate quality reports for regulatory compliance and export documentation.
 - Pre-Conditions: Inspection data is available, and the system is connected to the database.

- Post-Conditions: A detailed report (e.g., PDF or CSV) is exported, meeting SLSI and international standards (e.g., Codex Alimentarius).
- Use Case: Trigger Automated Batch Rejection
 - Actor: Saltern Operator
 - Description: The system detects a non-compliant batch (e.g., high impurity levels) and automatically activates mechanical actuators to remove it from the production line.
 - Pre-Conditions: Actuators are integrated with the system, and thresholds for rejection are set.
 - Post-Conditions: Non-compliant batch is isolated, and an alert is logged for operator review.
- Use Case: Validate and Optimize Model Performance
 - Actor: Quality Control Supervisor
 - Description: The supervisor conducts trials in a live environment, compares model decisions with human evaluations, and initiates model retraining to improve accuracy and latency.
 - Pre-Conditions: Trial data is collected, and the CNN model is accessible for updates.
 - Post-Conditions: Model accuracy exceeds 95%, and updated model is deployed with improved performance metrics.
- Use Case: Monitor Export Compliance
 - Actor: Salt Exporter
 - Description: The exporter monitors real-time purity levels and grading via the web platform to ensure compliance with international standards, adjusting parameters if needed.
 - Pre-Conditions: System is online, and export thresholds are configured.
 - Post-Conditions: Compliance certificate is generated, and batch rejection rates are below 5%.

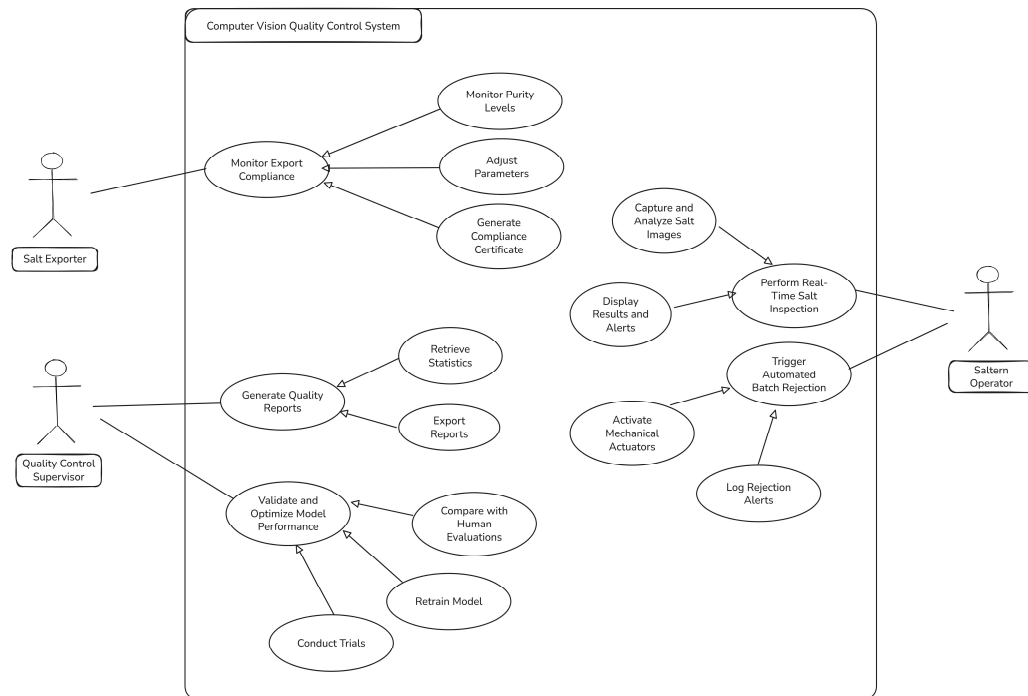


Figure VI: Use case Diagram

4.6. Test Cases

- Test Case ID: TC-001 - Image Acquisition and Processing
 - Description: Verify that the system captures and processes high-resolution images of salt samples in real-time
 - Pre-Conditions: 2K webcam is connected, LED ring lights are active, and conveyor simulation is running.
 - Test Steps:
 - Start the system and initiate live video streaming.
 - Place salt samples with varying impurities on the conveyor.
 - Capture 10 frames and process them through the CNN model.
 - Expected Result: All frames are captured with clear resolution, processed in <500ms per frame, and impurity detection is initiated.
- Test Case ID: TC-002 - Impurity Detection Accuracy
 - Description: Validate the CNN model's ability to detect impurities (e.g., clay, sand, black particles) accurately.

- Pre-Conditions: Trained CNN model is deployed, and annotated dataset with impurities is available.
- Test Steps:
 - Input 50 annotated images with known impurities.
 - Run the model to classify impurities vs. clean salt.
 - Compare results with ground truth labels.
- Expected Result: Accuracy $\geq 95\%$, with precision, recall, and F1-score > 0.9 for impurity detection.
- Test Case ID: TC-003 - Multi-Parameter Grading
 - Description: Test the system's ability to grade salt based on whiteness, size, and uniformity in real-time.
 - Pre-Conditions: System is online, and diverse salt samples are prepared.
 - Test Steps:
 - Capture images of salt samples with varying whiteness and crystal sizes.
 - Process images through the CNN to assign grades.
 - Review grading distribution on the dashboard.
 - Expected Result: Grading matches expert evaluation within 5% variance, displayed in < 1 second.
- Test Case ID: TC-004 - Automated Batch Rejection
 - Description: Verify integration with mechanical actuators to reject non-compliant batches.
 - Pre-Conditions: Actuator is connected, and rejection threshold is set (e.g., $> 10\%$ impurities).
 - Test Steps:
 - Introduce a batch with 15% impurities on the conveyor.
 - Monitor system response and actuator activation.
 - Check alert logs for rejection event.
 - Expected Result: Actuator removes the batch within 1 second of detection, and an alert is logged.

- Test Case ID: TC-005 - Dashboard Usability
 - Description: Ensure the user interface displays real-time results and alerts effectively.
 - Pre-Conditions: Web platform is accessible, and operator is logged in.
 - Test Steps:
 - Start a live inspection session.
 - Verify display of results, statistics, and alerts for a non-compliant batch.
 - Export a sample report.
 - Expected Result: Dashboard loads in <2 seconds, updates every 500ms, and report exports successfully in PDF format.
- Test Case ID: TC-006 - Performance Under Load
 - Description: Test system performance with continuous operation and high data volume.
 - Pre-Conditions: System is running, and 1000 images are queued for processing.
 - Test Steps:
 - Process all images in a simulated 1-hour session.
 - Measure latency and throughput.
 - Expected Result: Latency remains <500ms per frame, and throughput exceeds 30 frames/second.
- Test Case ID: TC-007 - Model Retraining Validation
 - Description: Confirm the system can retrain the CNN model with new data to improve accuracy.
 - Pre-Conditions: New dataset with unseen impurities is prepared.
 - Test Steps:
 - Upload new dataset and initiate retraining.
 - Test model with 20 new images post-retraining.
 - Expected Result: Retraining completes in <48 hours, and accuracy improves by at least 2% on new data.

4.7. Wireframes



Figure VII: Dashboard Section Wireframe

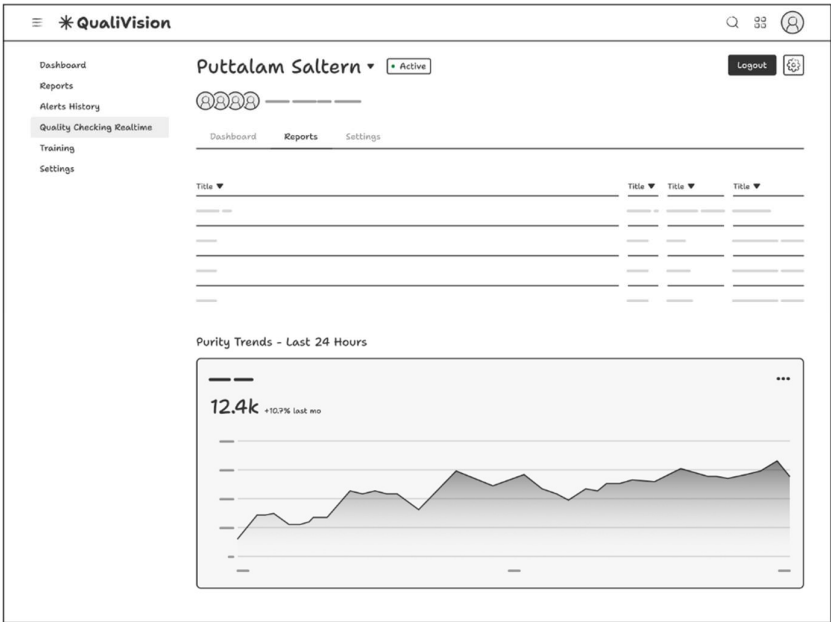


Figure VIII: Purity Section Wireframe

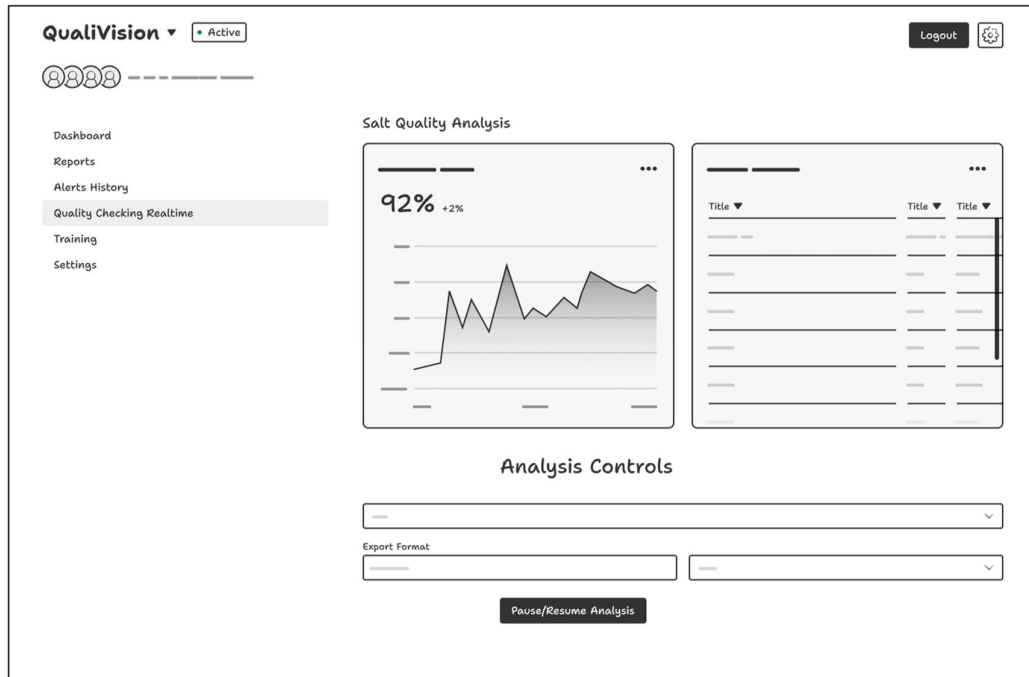


Figure IX: Realtime Quality Inspect Section Wireframe

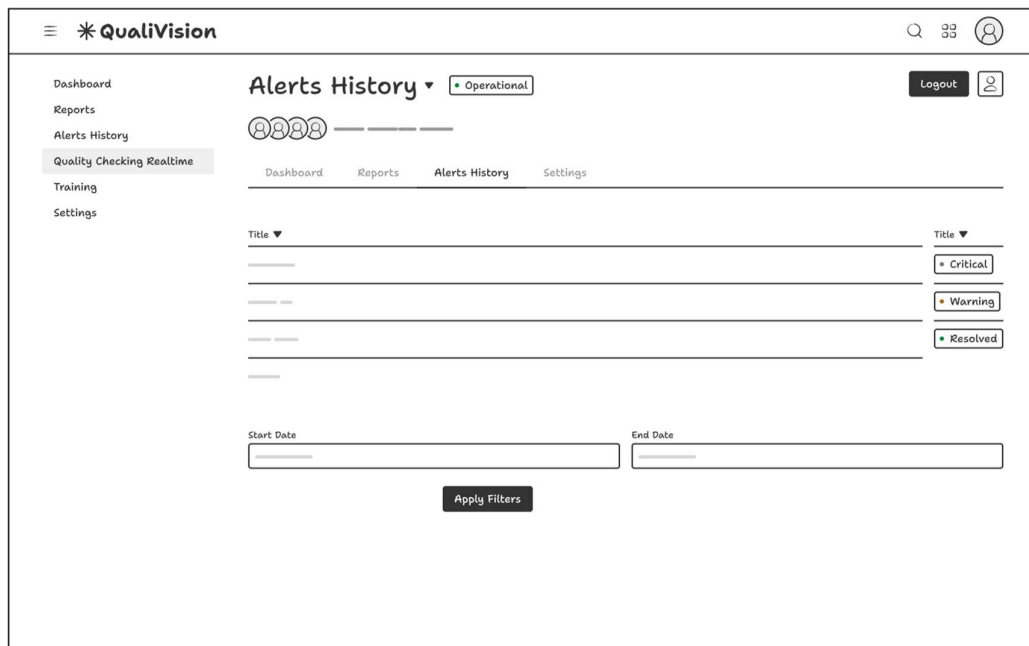


Figure X: Alert Section Wireframe

5. BUDGET AND BUDGET JUSTIFICATION

Expense	Cost (LKR)	Net Total (LKR)	Justification
Personal Costs			
Travelling	36,000	36,000	3000 x 12 covering site visits to Puttalam and supervisor meetings.
Software And Service			
Domain Purchase	1,000		4000 / 4 mem
Host and deploy	17,500		AWS EC2 Instance (200\$) 70,000/4
Perplexity Pro	18,000		6000 x 12 / 4
Premium Api	12,500		For data gathering (e.g.: weather) 50,000 / 4
Google Collab Pro	36,000		For model training (10\$ = 3000 per month)
Cloud Storage	20,000	105,000	Additional storage if needed
Grand Total		141,000	

Table 2: Budget Allocation for Research

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7. APPENDICES